

Course Overview

Stat 151A: Linear Models

- Unit 0
 - Linear algebra
 - * Formal operations: Valid dimensions in matrix multiplication, transpose, trace
 - * Algebra: As a representation of linear systems, invertibility, determinant
 - * Geometry: orthogonal vectors, basis vectors, linear combinations of basis vectors
 - * Eigenvalues and eigenvectors of square symmetric matrices
 - Probability
 - * Means, covariance, variance, correlation
 - * Conditional expectation, conditional variance, independence
 - * Asymptotics: Law of large numbers, central limit theorem, continuous mapping theorem
 - * The standard normal distribution
 - Tasks in data analysis
 - * Descriptive statistics
 - * Prediction
 - * Inference
- Unit 1
 - Simple linear regression
 - * The basic formula
 - * Sample means as a special case
 - * Coefficient as a measure of association
 - * Asymmetry between x and y
 - Matrix form
 - * The regressor matrix and response and residual vectors
 - * The derivation of the formula for vector-valued β
 - * Being able to express familiar simpler regressions in matrix form
 - Including sample mean, simple linear regression

- * ESS, RSS, TSS, and R squared, and the limitations of these measures of fit
- Interpretations
 - * As a descriptive statistic
 - * As a maximum likelihood estimator under normality
 - * As a consistent estimator of a true coefficient
 - * As a risk minimizer for prediction
- Transformations
 - * One-hot encodings and categorical variables
 - * Linear transformations of regressors leave the fit unchanged
 - A one-hot encoding and a constant
 - Rescaling and centering continuous variables
 - * Non-linear transformations of regressors give more expressive regressions
 - Polynomials, indicators, splines
 - * Interactions between regressors and indicator variables
 - Example: different slopes in different categories
 - * Transformations of the response
 - Changing units of the response
 - Choosing the scale that makes the most sense
 - Log transformations of heavy tailed responses
- Unit 2
 - The logic of testing confidence intervals in general
 - The selection of rejection regions
 - Level and power
 - Nulls and alternatives
 - Statistical tests
 - * The multivariate normal and its linear transformations
 - * How to construct tests of linear combinations of the OLS coefficients with the t-statistic
 - * Testing single entries as linear combinations as t-statistics
 - * Tests of multiple linear combinations of OLS coefficients via the F statistic
 - * Tests that sets of coefficients are zero as F statistics
 - Four classes of assumptions, when they might apply, and what they imply
 - * Normal errors
 - Assumed “true” parameters
 - Rarely applies without good reason
 - Distribution of OLS given in closed form for all N for fixed regressors
 - Only assumption under which we have chi2, t distributions and F distributions
 - * Homoskedastic errors

- Assumed “true” parameters
 - Residuals have the same variance, which is more plausible than normality but often suspect
 - Similar to the normal, but only asymptotically via the CLT
 - * Heteroskedastic errors
 - Assumed “true” parameters
 - Residual variance can depend on x
 - Similar to the normal asymptotically, but with different “sandwich” covariance
 - * Machine learning assumption
 - IID pairs (x, y)
 - More appropriate for prediction problems, though IID assumption still needs to be taken critically
 - No “true” parameters, but there is a limiting value as N goes to infinity that plays the role of true parameters
 - Inference is the same as in heteroskedastic errors
- Unit 3
 - Expressive basis functions
 - * What are some examples? (Splines, polynomials)
 - * Why do we use them in a prediction context?
 - * What are the tradeoffs determining how many we should use?
 - The bias-variance tradeoff
 - * In the context of linear regression, what does this mean? (Squared error, conditional on the regressors)
 - * Why is it important for prediction problems?
 - * What are some ways to trade off bias for variance in regression problems.
 - Ridge regression
 - * What is it? (Formula, Bayesian interpretation)
 - * How does the ridge parameter affect the bias and variance in the bias-variance tradeoff?
 - * Why do we use ridge regression together with expressive basis functions?
 - Unit 4
 - FWL theorem
 - * Be able to apply it with guidance (as in the homework)
 - * Understand what it means for constants (centering regressors)
 - Errors in regressors / regression to the mean
 - * Recognizing that inference with OLS assumes that x is measured without noise
 - * If you have noise in your regressors, you underestimate the true effect exactly as in ridge regression

- * Be able to recognize examples where you're actually trying to measure the effect of something you only observe with noise
- Omitted variables
 - * Why do we care for inference? Why don't we care for prediction?
 - * When is omitted variable bias a problem? (When the omitted variable is associated with both the response and variable of interest — apply the FWL theorem)
 - * Understand why you can't get around omitted variables by putting every possible regressor in the regression
- Influence and outliers
 - * Whether something is an outlier depends strongly on context
 - * Understand the different kinds of influence of x outliers and y outliers
 - * Know how to diagnose (residuals for y, leverage for x)
 - * Know basic properties of leverage scores (between 0 and 1, sum to P)